VARIOUS TYPES OF ACTIVATION FUNCTIONS AND TERMINOLOGY OF NEURAL NETWORK

Introduction

A Neural Network is a machine learning model inspired by the way the human brain works. It is made up of interconnected nodes called **neurons**, which process information and learn patterns from data.

Neural networks are widely used in applications such as image recognition, speech recognition, recommendation systems, medical diagnosis, natural language processing, and prediction.

Two important concepts in neural networks are **neural network terminology** and **activation functions**. Understanding the basic terms helps us understand how a neural network works, while activation functions help the network learn complex patterns.

Terminology of Neural Network

A neural network consists of several components. The important terms are explained below.

Neuron

A neuron is the basic unit of a neural network. It receives input values, performs a calculation, and produces an output.

A neuron takes inputs, multiplies them by weights, adds a bias, and then passes the result through an activation function.

Input

Input is the information given to a neural network.

For example, if a neural network is used to predict a student's result, the inputs could be:

- Study hours
- Attendance percentage
- Previous marks
- Assignment marks

Weight

A weight represents the importance of an input.

Each input is multiplied by a weight. During training, the neural network changes these weights to improve its predictions.

For example, if attendance is more important than another feature, the network may learn a larger weight for attendance.

Bias

Bias is an additional value added to the weighted sum of inputs. It helps the neuron adjust its output more effectively.

The basic calculation performed by a neuron is:

$$\mathbf{Z = (X_1W_1 + X_2W_2 + X_3W_3 + \dots) + b}$$

Where:

- X = Input
- W = Weight
- b = Bias
- Z = Weighted sum

Layer

A layer is a group of neurons working together.

There are mainly three types of layers:

Input Layer:
Receives the input data.

Hidden Layer:
Processes the input and learns patterns. A neural network can have one or many hidden layers.

Output Layer:
Produces the final result or prediction.

Input Layer

The input layer is the first layer of a neural network. It receives the features of the dataset.

For example, if a dataset has three features, the input layer may contain three input nodes.

Hidden Layer

Hidden layers are located between the input and output layers. They perform calculations and identify patterns in the data.

A neural network can contain one or more hidden layers.

Output Layer

The output layer is the final layer of the neural network. It provides the result.

For example:

- Yes/No → Classification
- Cat/Dog → Classification
- House price → Regression

Forward Propagation

Forward propagation is the process of passing input data through the neural network from the input layer to the output layer.

The network calculates the weighted sum and applies activation functions at each layer to produce an output.

Loss Function

A loss function measures the difference between the actual output and the predicted output.

A smaller loss means that the prediction is closer to the actual value.

Examples:

- Mean Squared Error (MSE)
- Cross-Entropy Loss

Backpropagation

Backpropagation is a training method used to reduce the error of a neural network.

It calculates how much each weight contributed to the error and adjusts the weights so that the network can make better predictions.

Epoch

An epoch means one complete pass through the entire training dataset.

For example, if a model is trained for 10 epochs, the complete training dataset is processed 10 times.

Batch Size

Batch size is the number of training samples processed at one time before updating the weights.

For example, if the batch size is 32, the model processes 32 samples before updating the weights.

Learning Rate

Learning rate controls how much the weights are changed during training.

- A very small learning rate can make training slow.
- A very large learning rate may cause the model to miss the best solution.

Training

Training is the process of teaching a neural network using a dataset.

During training, the network adjusts its weights and biases to reduce the prediction error.

Prediction

Prediction is the output produced by the trained neural network when new data is given to it.

Activation Function

An **activation function** is a mathematical function used in a neural network to determine the output of a neuron.

The main purpose of an activation function is to introduce **non-linearity** into the neural network.

Without activation functions, a neural network would behave like a simple linear model and would have difficulty learning complex patterns.

The general process is:

Input → Weighted Sum + Bias → Activation Function → Output

Activation functions are especially important in hidden layers and the output layer.

Types of Activation Functions

There are several types of activation functions used in neural networks.

Binary Step Function

The Binary Step Function produces either 0 or 1 depending on the value of the input.

Formula:

$f(x) = 1$, if $x \geq 0$

$f(x) = 0$, if $x < 0$

Example:

If $x = 5$, output = 1

If $x = -2$, output = 0

Advantages:

- Very simple
- Easy to understand

Disadvantages:

- Not suitable for modern deep neural networks
- Gradient is not useful for learning

Use:

It can be used for simple binary decision problems.

Linear Activation Function

The linear activation function produces an output directly proportional to its input.

Formula:

$f(x) = x$

Example:

If $x = 5$, output = 5

If $x = -3$, output = -3

Advantages:

- Very simple
- Useful for regression problems

Disadvantages:

- Cannot learn complex non-linear patterns

- Usually not suitable for hidden layers

Use:

It is commonly used in the output layer of regression models.

Sigmoid Function

The Sigmoid function converts values into a range between **0 and 1**.

Formula:

$f(x) = 1 / (1 + e^{-x})$

Example:

For a large positive value, the output approaches 1.

For a large negative value, the output approaches 0.

Advantages:

- Output is between 0 and 1
- Useful for binary classification
- Easy to interpret as a probability

Disadvantages:

- Can suffer from the vanishing gradient problem
- Training can become slow for deep networks

Use:

Sigmoid is commonly used in the output layer for binary classification.

Tanh Function

Tanh stands for **Hyperbolic Tangent**.

It produces values between **-1 and +1**.

Formula:

$f(x) = \tanh(x)$

Advantages:

- Output is centered around zero
- Can work better than sigmoid in some hidden layers

Disadvantages:

- Can suffer from the vanishing gradient problem
- Less commonly used in modern deep networks compared with ReLU

Use:

It can be used in hidden layers and some recurrent neural networks.

ReLU Function

ReLU stands for **Rectified Linear Unit**.

It is one of the most commonly used activation functions in neural networks.

Formula:

f(x) = max(0, x)

This means:

- If x is positive, output = x
- If x is negative, output = 0

Example:

If x = 5, output = 5

If x = -3, output = 0

Advantages:

- Simple and fast
- Helps neural networks learn faster
- Reduces the vanishing gradient problem for positive inputs

Disadvantages:

- Can suffer from the **Dying ReLU** problem, where some neurons may stop responding

Use:

ReLU is widely used in hidden layers of deep neural networks.

Leaky ReLU

Leaky ReLU is a modified version of ReLU. Instead of making all negative values zero, it allows a small negative value.

Formula:

$f(x) = x$, if $x > 0$

$f(x) = \alpha x$, if $x \leq 0$

Here, α is a small value such as 0.01.

Example:

If $x = 5$, output = 5

If $x = -5$, output = -0.05

Advantages:

- Helps reduce the Dying ReLU problem
- Allows a small gradient for negative values

Disadvantages:

- Requires choosing a value for α
- Not always better than ReLU

ELU

ELU stands for **Exponential Linear Unit**.

It behaves like ReLU for positive values but produces a smooth negative output for negative values.

Advantages:

- Can reduce the Dying ReLU problem
- Provides smoother outputs for negative values

Disadvantages:

- More computationally expensive than ReLU

Softmax Function

Softmax converts a group of values into probabilities whose total is equal to 1.

It is mainly used for **multi-class classification**.

Example:

Suppose a model classifies an image into three categories:

- Cat → 0.70
- Dog → 0.20
- Bird → 0.10

The total is:

0.70 + 0.20 + 0.10 = 1.00

The class with the highest probability is selected as the prediction.

Advantages:

- Produces probabilities
- Useful for multi-class classification

Use:

Softmax is commonly used in the output layer when there are multiple classes.

Comparison of Activation Functions

Activation Function	Output Range	Common Use
Binary Step	0 or 1	Simple binary decisions
Linear	$-\infty$ to $+\infty$	Regression
Sigmoid	0 to 1	Binary classification
Tanh	-1 to +1	Hidden layers, some RNNs
ReLU	0 to $+\infty$	Hidden layers
Leaky ReLU	$-\infty$ to $+\infty$	Hidden layers
ELU	$-\alpha$ to $+\infty$	Hidden layers
Softmax	0 to 1	Multi-class classification

Importance of Activation Functions

Activation functions are important because they allow neural networks to learn complex relationships in data.

Their main purposes are:

1. They introduce non-linearity into the network.
2. They help neurons produce useful outputs.
3. They allow neural networks to learn complex patterns.
4. They help determine the final output of neurons.
5. Different activation functions are suitable for different types of problems.

For example, **ReLU** is commonly used in hidden layers, **Sigmoid** can be used for binary classification, and **Softmax** is commonly used for multi-class classification.

Simple Example of a Neural Network

Consider a neural network used to predict whether a student will pass or fail.

Inputs:

- Study Hours
- Attendance
- Assignment Marks

These values are given to the **input layer**.

The input values are multiplied by **weights** and a **bias** is added. The result is passed through an **activation function** in the hidden layer.

The processed information then reaches the **output layer**.

The output may be:

1 → Pass

0 → Fail

During training, the neural network compares its prediction with the actual result and adjusts the weights using backpropagation.

Applications of Neural Networks

Neural networks are used in many areas, such as:

- Image recognition
- Face recognition
- Speech recognition
- Chatbots
- Recommendation systems
- Medical diagnosis
- Fraud detection
- Weather prediction
- Stock market prediction
- Natural Language Processing
- Autonomous vehicles

Conclusion

Neural networks are an important part of modern machine learning and artificial intelligence. They consist of different components such as neurons, weights, biases, layers, loss functions, and learning rates.

Activation functions play an important role in neural networks because they introduce non-linearity and allow the model to learn complex patterns. Different activation functions are used for different purposes. ReLU is commonly used in hidden layers, Sigmoid is useful for binary classification, Linear is suitable for regression, and Softmax is commonly used for multi-class classification.

Understanding neural network terminology and activation functions provides a basic foundation for learning more advanced topics in deep learning and artificial intelligence.